

Sibley School of Mechanical and Aerospace Engineering

MAE 3240 Heat Transfer

Spring 2026

Problem Set #1

Posted: Monday, Jan 26

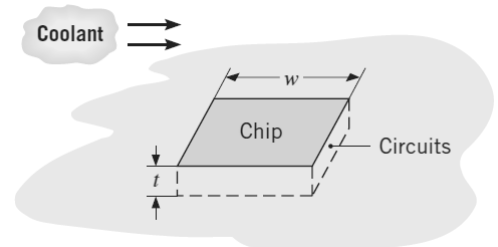
Due: Tuesday, Feb 3, 11:59 pm

1. (30 pt) Consider a spacecraft traveling in the space *outside* the atmosphere, discuss the dominant heat transfer mode across the spacecraft and briefly explain. After completing the space exploration mission, this spacecraft returns to earth by traveling *through* the atmosphere. Please discuss the dominant heat transfer mode across the spacecraft and briefly explain.



Note: you do not need to follow the recommended format for this problem, since this is a short-answer problem.

2. (30 pt) A square silicon chip ($k = 150 \text{ W}/(\text{m}\cdot\text{K})$) is of width $w = 5 \text{ mm}$ on a side and of thickness $t = 1 \text{ mm}$. The chip is mounted in a substrate such that its side and back surface are thermally insulated. The front surface is exposed to a coolant with convective flows. The chip dissipates 9 W energy under steady-state operation. Assume that thermal radiation is negligible in this problem. The coolant temperature is $T_\infty = 20^\circ\text{C}$.



- (a) What is the steady-state temperature difference between back and front surfaces of the chip?
- (b) If the front surface temperature of the chip cannot be above 80°C , calculate the minimum heat transfer coefficient required for the coolant flow. Compare your calculation with heat transfer coefficients shown in Table 1.1 of the textbook and discuss what convection processes can be used to cool this chip.

Note → solar vs surrounding

3. (40 pt) The emissivity of a roof material is $\varepsilon = 0.18$. The absorptivity of this material for solar irradiation is $\alpha = 0.76$.

(a) Consider a day of solar irradiation $G = 750 \text{ W/m}^2$, ambient temperature $T_\infty = 300 \text{ K}$, surrounding temperature $T_{\text{surr}} = 300 \text{ K}$, and heat transfer coefficient of air flow $h = 10 \text{ W/(m}^2 \cdot \text{K)}$, calculate the temperature of the roof. Assume the bottom surface of the roof is thermally insulated. Would the neighboring cat be comfortable walking on the roof constructed by this material?



(b) According to your calculation, discuss the dominant heat transfer mode in (a).

Cover Page

Name: Julie Nam

Problem set number: #1

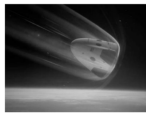
Date of Submission: 2/3/2026

Due date: 2/3/2026

Acknowledgement:

- I used lecture notes
- I went to office hours (Monday around 6:40 pm)
- Compared solutions with Charles Valembrun

1. (30 pt) Consider a spacecraft traveling in the space *outside* the atmosphere, discuss the dominant heat transfer mode across the spacecraft and briefly explain. After completing the space exploration mission, this spacecraft returns to earth by traveling *through* the atmosphere. Please discuss the dominant heat transfer mode across the spacecraft and briefly explain.

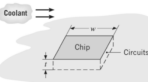


Note: you do not need to follow the recommended format for this problem, since this is a short-answer problem.

Outside the atmosphere, the dominant heat transfer mode across the spacecraft would be radiation, especially from the sun. Since space is a vacuum with no air or medium, there won't be any conduction or convection.

When the spacecraft returns to earth by traveling through the atmosphere, the dominant heat transfer mode would be ★ convection. This time, there is air surrounding the spacecraft, and as it moves at high speeds, heat flows through convection from hot gases to the surface of aircraft.

2. (30 pt) A square silicon chip ($k = 150 \text{ W/(m}\cdot\text{K)}$) is of width $w = 5 \text{ mm}$ on a side and of thickness $t = 1 \text{ mm}$. The chip is mounted in a substrate such that its side and back surface are thermally insulated. The front surface is exposed to a coolant with convective flows. The chip dissipates 9 W energy under steady-



state operation. Assume that thermal radiation is negligible in this problem. The coolant temperature is $T_\infty = 20^\circ\text{C}$.

- (a) What is the steady-state temperature difference between back and front surfaces of the chip?
 (b) If the front surface temperature of the chip cannot be above 80°C , calculate the minimum heat transfer coefficient required for the coolant flow. Compare your calculation with heat transfer coefficients shown in Table 1.1 of the textbook and discuss what convection processes can be used to cool this chip.

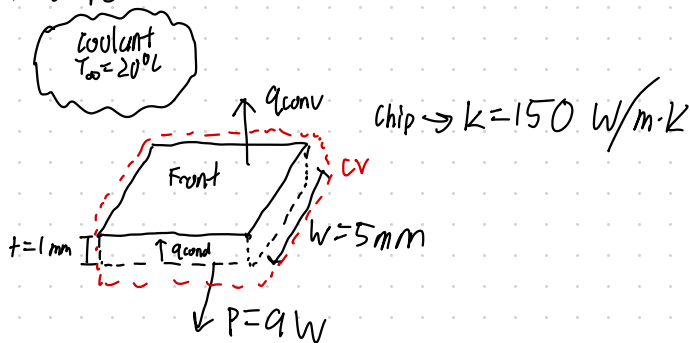
Known: $k = 150 \text{ W/m}\cdot\text{K}$

- $w = 5 \text{ mm}$
- $t = 1 \text{ mm}$
- side and back insulated
- Front $\rightarrow$ convective
- Dissipates 9 W under steady state
- Radiation negligible
- Coolant $\rightarrow T_\infty = 20^\circ\text{C}$

square silicon chip

Find: (a) steady-state temp. diff. between front and back surfaces of chip
 (b) Front temp $\rightarrow$ can't be above $80^\circ\text{C} \rightarrow$ Find min. heat transfer coefficient required for coolant flow
 • compare with table 1.1
 • what convection processes can cool chip?

Schematic:



Assumptions: - steady-state
 • constant properties
 • negligible thermal radiation
 • side and back perfectly insulated
 • negligible ΔKE and ΔPE of air flow

• coolant $\rightarrow$ convective flow
 • 1D flow

Analysis: (a) Front surface $\rightarrow q_x = -kA \frac{dT}{dx}$

$$q_x = \frac{-kA \Delta T}{t}$$

$$\Delta T = \frac{q_{\text{ext}}}{kA}$$

$$\frac{(9 \text{ W})(0.001 \text{ m})}{150 \text{ W/m}\cdot\text{K} \cdot 0.005 \cdot 0.005 \text{ m}^2}$$

$$\Delta T = 2.4 \text{ K}$$

$$\dot{E}_{\text{in}} - \dot{E}_{\text{out}} = 0$$

$$q_{\text{cond}} - q_{\text{conv}} = 0$$

$$(b) \quad q = hA(T_s - T_{\infty})$$

$$h = \frac{q}{A(T_s - T_{\infty})}$$

$$h = \frac{9 \text{ W}}{0.005^2 \text{ m}^2 (353 - 293 \text{ K})}$$

$$h = 6000 \frac{\text{W}}{\text{m}^2 \cdot \text{K}}$$

Since $h = 6000 \text{ W/(m}^2 \cdot \text{K)}$, forced convection could be used to cool this chip (suitable for liquids with $h = 100 - 20,000 \text{ W/(m}^2 \cdot \text{K})$)

3. (40 pt) The emissivity of a roof material is $\epsilon = 0.18$. The absorptivity of this material for solar irradiation is $\alpha_s = 0.76$.

- (a) Consider a day of solar irradiation $G_s = 750 \text{ W/m}^2$, ambient temperature $T_\infty = 300 \text{ K}$, surrounding temperature $T_{\text{sur}} = 300 \text{ K}$, and heat transfer coefficient of air flow $h = 10 \text{ W/(m}^2 \cdot \text{K)}$, calculate the temperature of the roof. Assume the bottom surface of the roof is thermally insulated. Assume the roof can be treated as



a grey body to the irradiation of surrounding environment. Would the neighboring cat be comfortable walking on the roof constructed by this material?

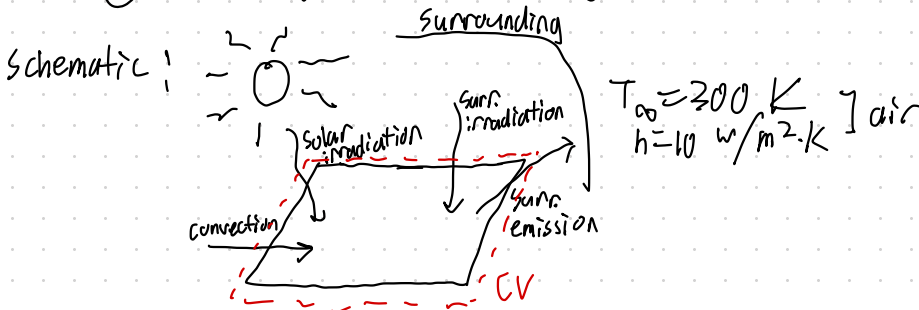
- (b) According to your calculation, discuss the dominant heat transfer mode in (a). $\rightarrow$ Fastest way to cool down

Known: emissivity $\epsilon = 0.18$
absorptivity $\alpha = 0.76$ } roof material

$$\begin{aligned} G_s &= 750 \text{ W/m}^2 \\ T_\infty &= 300 \text{ K} \\ T_{\text{sur}} &= 300 \text{ K} \\ h &= 10 \text{ W/(m}^2 \cdot \text{K)} \end{aligned}$$

Find: (a) Temp of roof
Would a cat be comfortable walking on roof?

(b) dominant heat transfer mode



- Assumptions:
- Roof $\rightarrow$ grey body
 - Bottom surface of roof is thermally insulated
 - steady state
 - constant air properties
 - Negligible DPE and AFE of air flow
 - Negligible work done by cat
 - 1D Flow

Analysis: (a) $q_{\text{rad}}'' = \epsilon \sigma (T_s^4 - T_{\text{sur}}^4)$ $\rightarrow$ surrounding
 $q_{\text{conv}}'' = h (T_s - T_\infty)$
 $q_{\text{rad}}'' = \alpha G_s$

steady state

$$\hookrightarrow q_{\text{in}} = q_{\text{out}}$$

$$\alpha G_s + \epsilon \sigma T_{\text{sur}}^4 = \epsilon \sigma T_s^4 + h (T_s - T_\infty)$$

$$(0.76)(750 \frac{\text{W}}{\text{m}^2}) + (0.18)(5.67 \times 10^{-8} \text{ W/(m}^2 \cdot \text{K}^4))(300 \text{ K})^4 = (0.18)(5.67 \times 10^{-8} \text{ W/(m}^2 \cdot \text{K}^4))T_s^4 + (10 \text{ W/(m}^2 \cdot \text{K)})(T_s - 300 \text{ K})$$

$$T_s \approx 350 \text{ K} \approx 77^\circ\text{C}$$

The cat would not be comfortable because a roof temperature of 77°C would be very too hot

(b) Radiation $\rightarrow \epsilon \sigma T_s^4$
 $(0.18)(5.67 \times 10^{-8} \text{ W/m}^2\text{K}^4)(350 \text{ K})^4$
 $q_{\text{rad, out}}'' = 153.15 \text{ W/m}^2$

Convection $\rightarrow h(T_s - T_\infty)$
 $(10 \text{ W/m}^2\text{K})(350 \text{ K} - 300 \text{ K})$
 $q_{\text{conv, out}}'' = 500 \text{ W/m}^2$

Comparing the heat transfer out of the roof through radiation vs. convection, convection is the dominant heat transfer mode